FINAL HOMEWORKS #7 & #8

HW #7 – Excel based- will post next week

(Portfolio Optimization & Bond Duration & Beta)

HW #8 – Calculus Review for the Final

(Pearson – now posted on Canvas)

BOTH will be due Thursday May 1st

FINAL Exam – Monday 5/5 (KOBL 100)***

Exam #2 - results

High: 100

Low: 50

Average: 76.5%

Yes- This class is difficult. But…

https://www.wsj.com/articles/top-colleges-high-paying-finance-jobs-e6742bb8

the average salaries that graduates earn in their first 10 years out of college:

Premium- higher pay than the median graduate in finance

Top 20 Public Colleges for Finance Salaries

These graduates earn higher pay than the median graduate in finance

RANK	COLLEGE	ANNUAL SALARY PREMIUM	FINANCE % OF GRADUATES	AVERAGE YEARLY SALARY
1	University of Virginia-Main Campus	$19,676	5.81%	$116,427
2	University of Michigan-Ann Arbor	$18,818	4.10%	$115,569
3	Binghamton University	$18,268	4.51%	$115,019
4	University of California-Berkeley	$18,040	2.89%	$114,791
5	William & Mary	$15,559	4.17%	$112,310
6	CUNY Bernard M Baruch College	$14,416	11.81%	$111,167
7	Rutgers University-New Brunswick	$14,386	4.47%	$111,137
8	Stony Brook University	$11,414	3.13%	$108,165
9	University of California-Los Angeles	$10,653	2.18%	$107,404
10	SUNY at Albany	$10,479	4.01%	$107,230
11	University of Utah	$10,296	4.22%	$107,047
12	Georgia Institute of Technology-Main Campus	$9,688	2.29%	$106,439
13	University of Texas at Austin	$8,969	3.15%	$105,720
14	University of Delaware	$8,521	5.41%	$105,272
15	University of Illinois Urbana-Champaign	$8,426	2.55%	$105,177
16	University of Connecticut	$6,725	2.69%	$103,476
17	University of Maryland-College Park	$6,711	3.01%	$103,462
18	Indiana University-Bloomington	$6,212	3.55%	$102,963
19	University of North Carolina at Chapel Hill	$5,788	5.37%	$102,539
20	University of Colorado Boulder	$5,500	2.50%	$102,251

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Grades

The following are the points assigned to the various graded components of this class:

Graded Activities	***Points Possible***
Exam 1	100
Exam 2	100
Exam 3 (Final Exam)	100
Homework	60
Quizzes	40
Total Possible Points	**400**

tentative course CURVE (see canvas)

THIS WILL *NOT* be the *exact* final curve to the course
***Subject to change… approximate but not absolute!**

tentative CURVE (see canvas)

Letter Grade	Range
A	100% to 90%
A-	< 90% to 88%
B+	< 88% to 85%
B	< 85% to 81%
B-	< 81% to 76%
C+	< 76% to 72%
C	< 72% to 69%
C-	< 69% to 65%
D+	< 65% to 60%
D	< 60% to 55%
D-	< 55% to 50%
F	< 50% to 0%

***Subject to change…**

approximate but not absolute!

THIS WILL *NOT* be the *exact* final curve to the course

Remaining Topics

Portfolio Optimization

Bond Duration

CAPM & Beta calculation (time permitting)

REVIEW:
Portfolio Expected Returns

The expected return of a portfolio is the weighted average of the expected returns for each asset in the portfolio:

$$E(R_P) = \sum_{i=1}^{n} w_i E(R_i)$$

w_i = % of portfolio invested in each asset

REVIEW:
Portfolio Standard Deviation

Two asset portfolio:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 Cov_{1,2}}$$

We used solver to minimize std. deviation…

Now we can confirm with calculus and derivatives!

REVIEW: Correlation

$$COR(X,Y) = \frac{COV(X,Y)}{\sigma_X \cdot \sigma_Y}$$

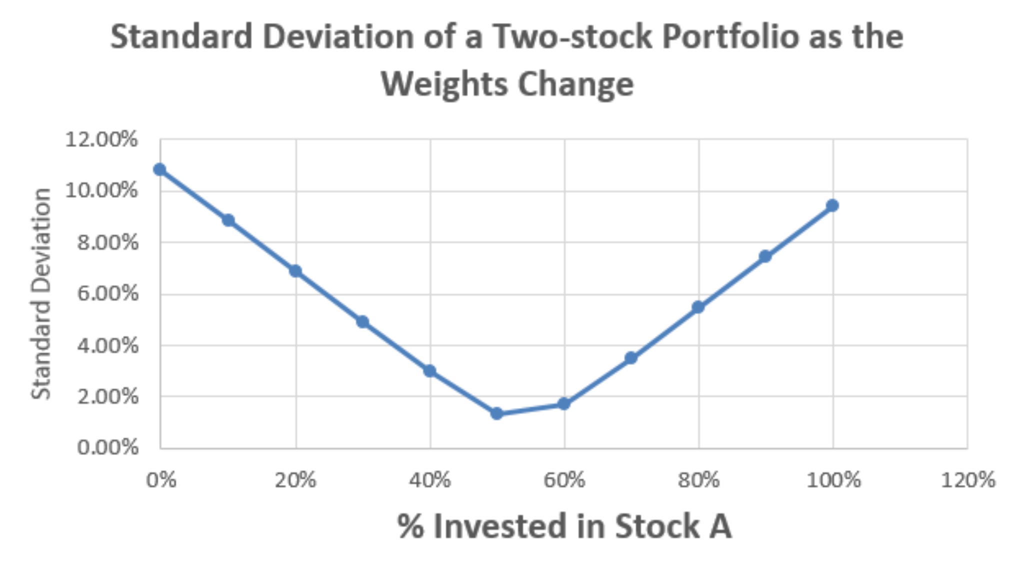
=correl(array1, array2)

Value between -1 to +1

Measure degree of linear relationship

Minimum Variance Portfolio

Weights		
Stock A	Stock B	Portfolio St. Dev.
100%	0%	9.43%
90%	10%	7.44%
80%	20%	5.46%
70%	30%	3.51%
60%	40%	1.74%
50%	50%	1.35%
40%	60%	2.95%
30%	70%	4.87%
20%	80%	6.84%
10%	90%	8.83%
0%	100%	10.83%



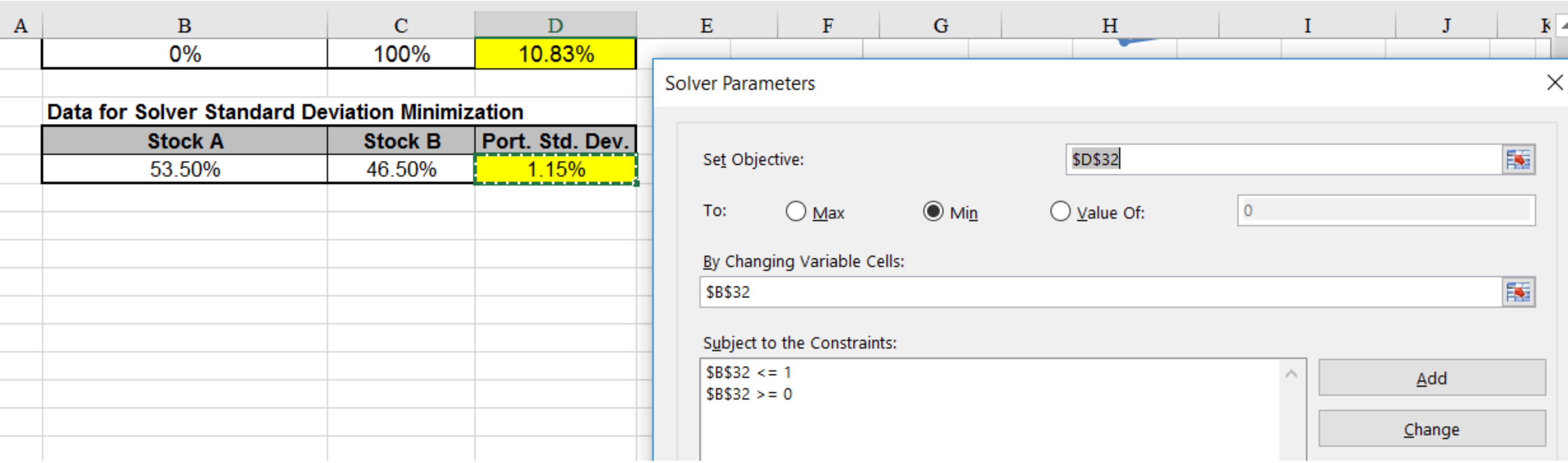
Some weighted combination of stocks A & B will provide a portfolio with minimized standard deviation

Minimum Variance Portfolio

solver:

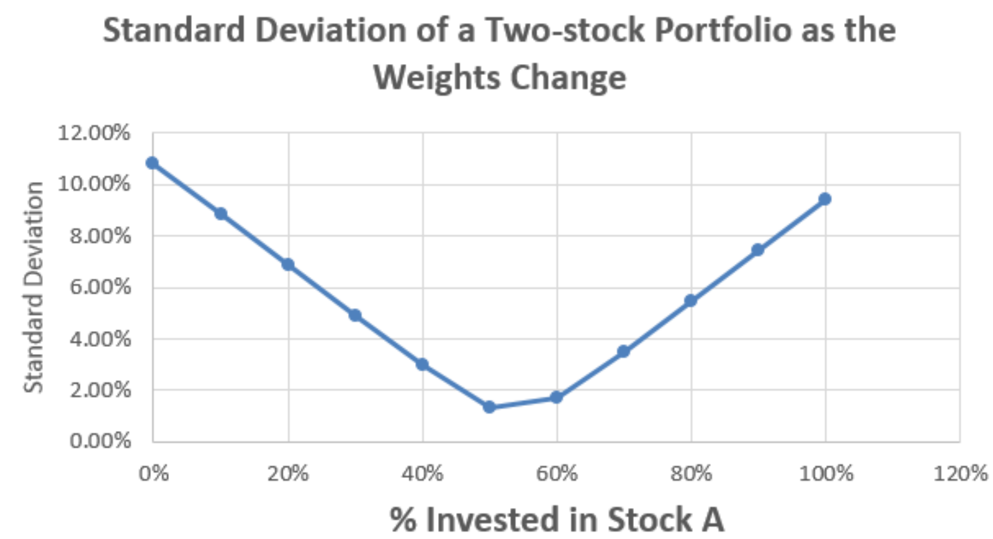
set "portfolio std. dev." to "min"

by changing "stock A weight"



Back to calculus:

Weights		
Stock A	Stock B	Portfolio St. Dev.
100%	0%	9.43%
90%	10%	7.44%
80%	20%	5.46%
70%	30%	3.51%
60%	40%	1.74%
50%	50%	1.35%
40%	60%	2.95%
30%	70%	4.87%
20%	80%	6.84%
10%	90%	8.83%
0%	100%	10.83%



When derivative of variance function is equal to zero:

$$W_A = \frac{\sigma_B^2 - Cov_{(A,B)}}{\sigma_A^2 + \sigma_B^2 - 2 * Cov_{(A,B)}}$$

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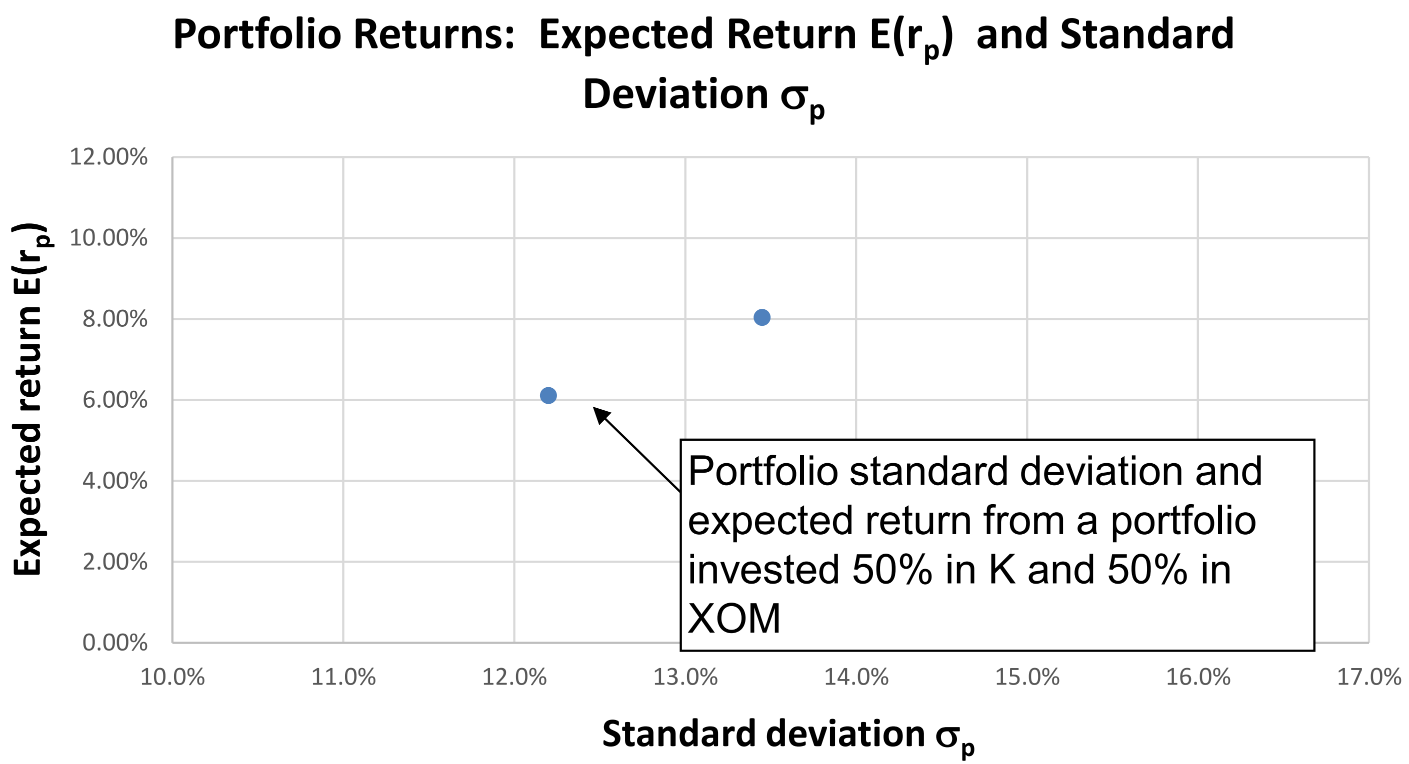
$$\sigma_p = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2Cov_{1,2}}$$

$$W_A = \frac{\sigma_B^2 - Cov_{(A,B)}}{\sigma_A^2 + \sigma_B^2 - 2 * Cov_{(A,B)}}$$

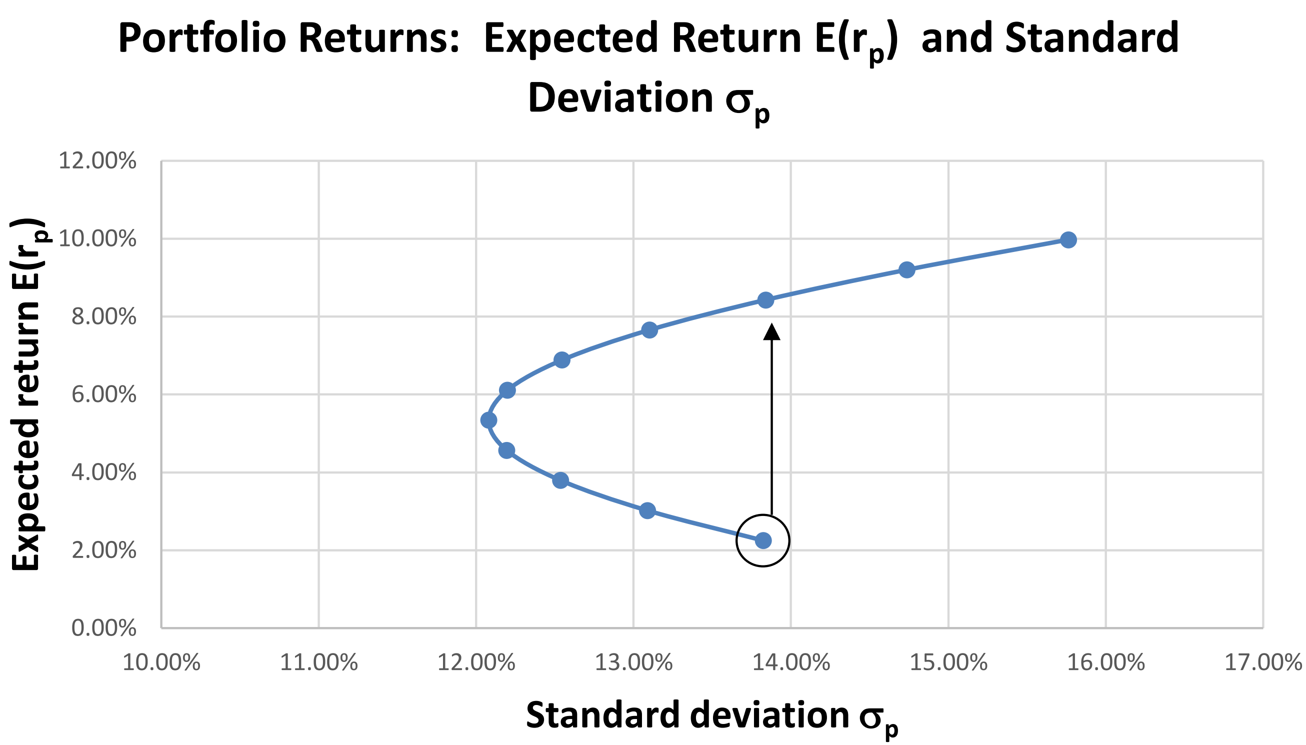
What about other portfolios?



We can generate expected return and standard deviation data for other alternatively weighted portfolios…

…and plot all the data

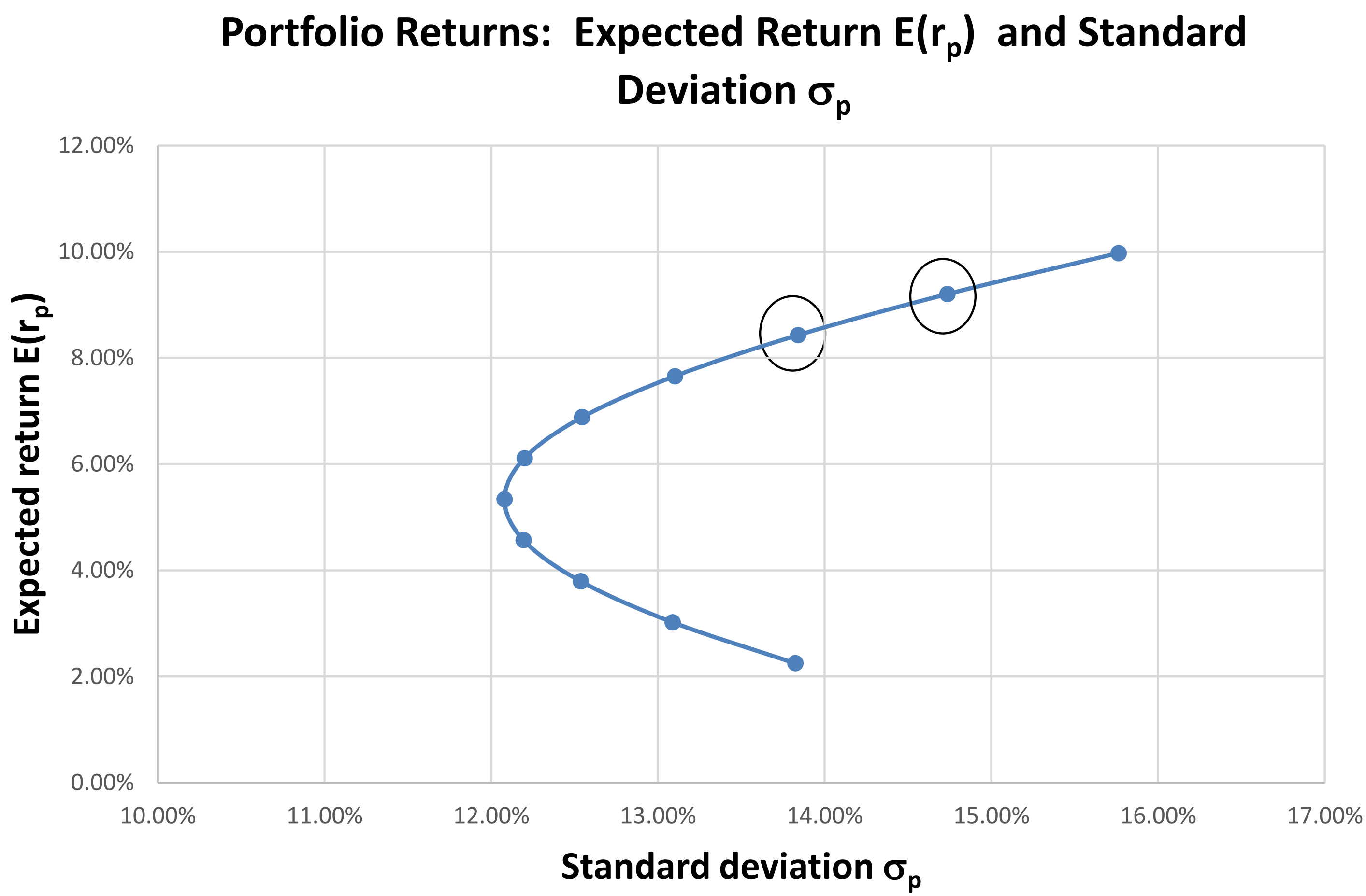
Expected Returns & Std. Deviation



The circled portfolio is not optimal –

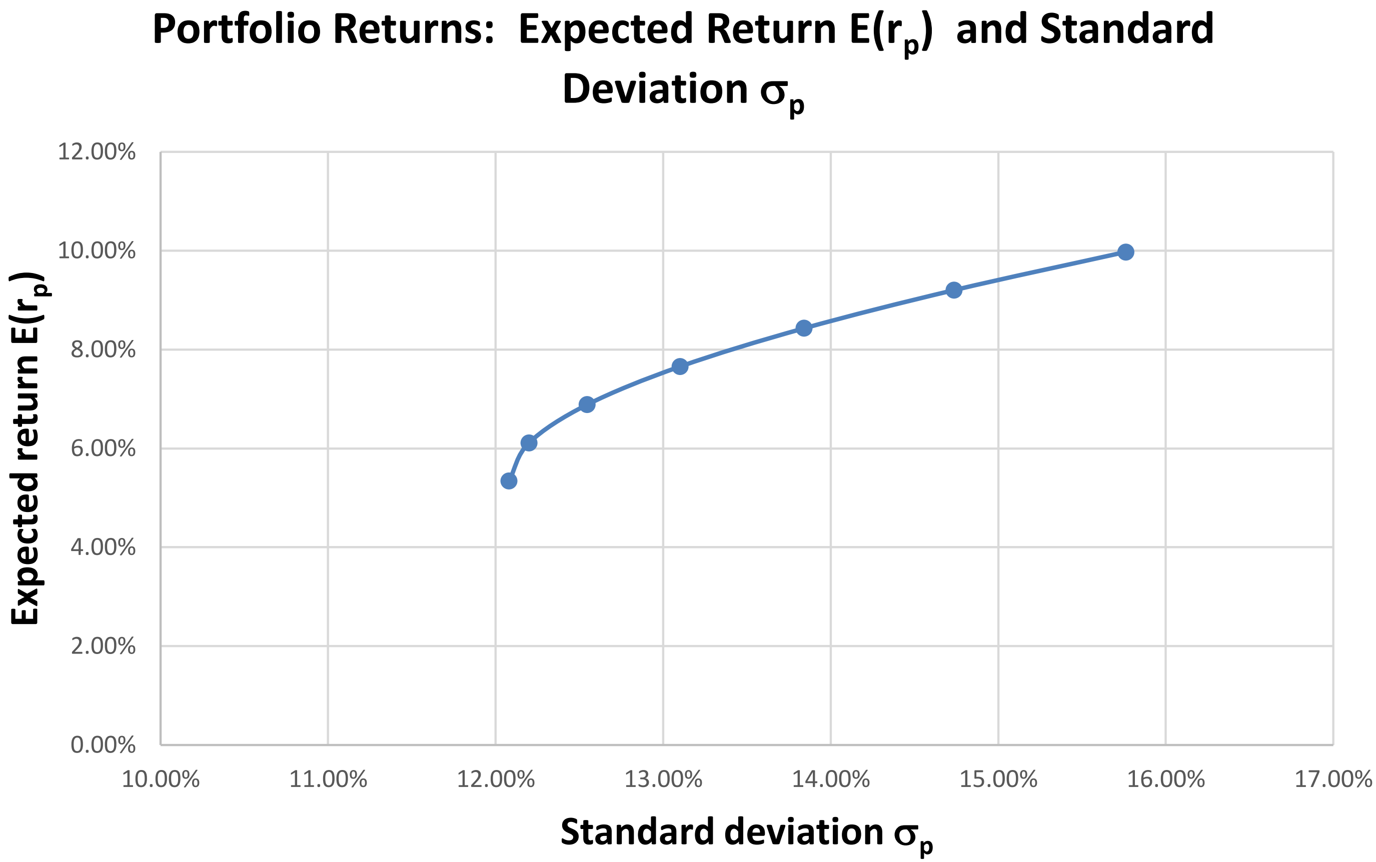
alternative portfolio has a higher expected return & the same std. deviation

Expected Returns & Std. Deviation



Risk/return trade-off between the two circled portfolios

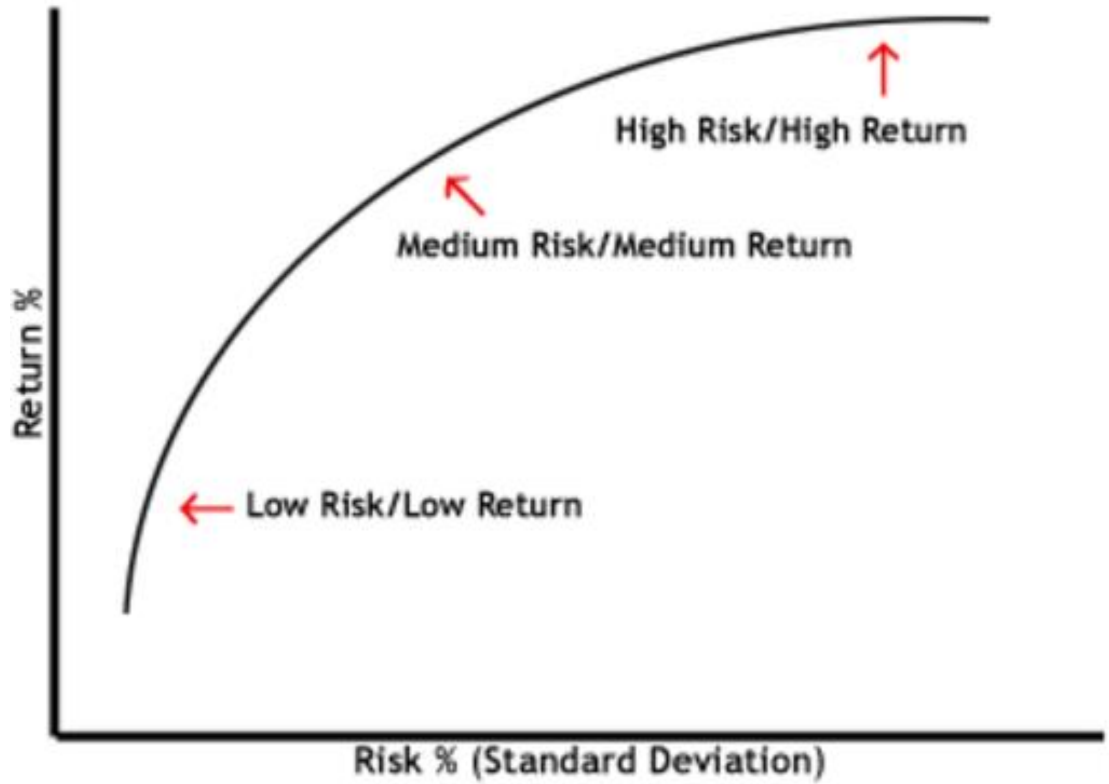
Portfolios – Efficient Frontier



Top part of the curve is the efficient frontier –

Set of those portfolios that offer the highest expected return for a given std. deviation

Efficient Frontier



Choice between any 2 portfolios on the efficient frontier involves a trade off between higher return and higher risk (std. deviation)

Effect of Correlation

With a correlation of +1, it is impossible to diversify away any risk

With a correlation between -1 and +1, some risk can be eliminated through diversification

With a correlation of -1, a risk free portfolio can be created using a combination of the two assets

Correlation

$$COR(X,Y) = \frac{COV(X,Y)}{\sigma_X \cdot \sigma_Y}$$

=correl(array1, array2)

Value between -1 to +1

Measure degree of linear relationship

EXCEL

***Open excel – EFFICIENT FRONTIER file**

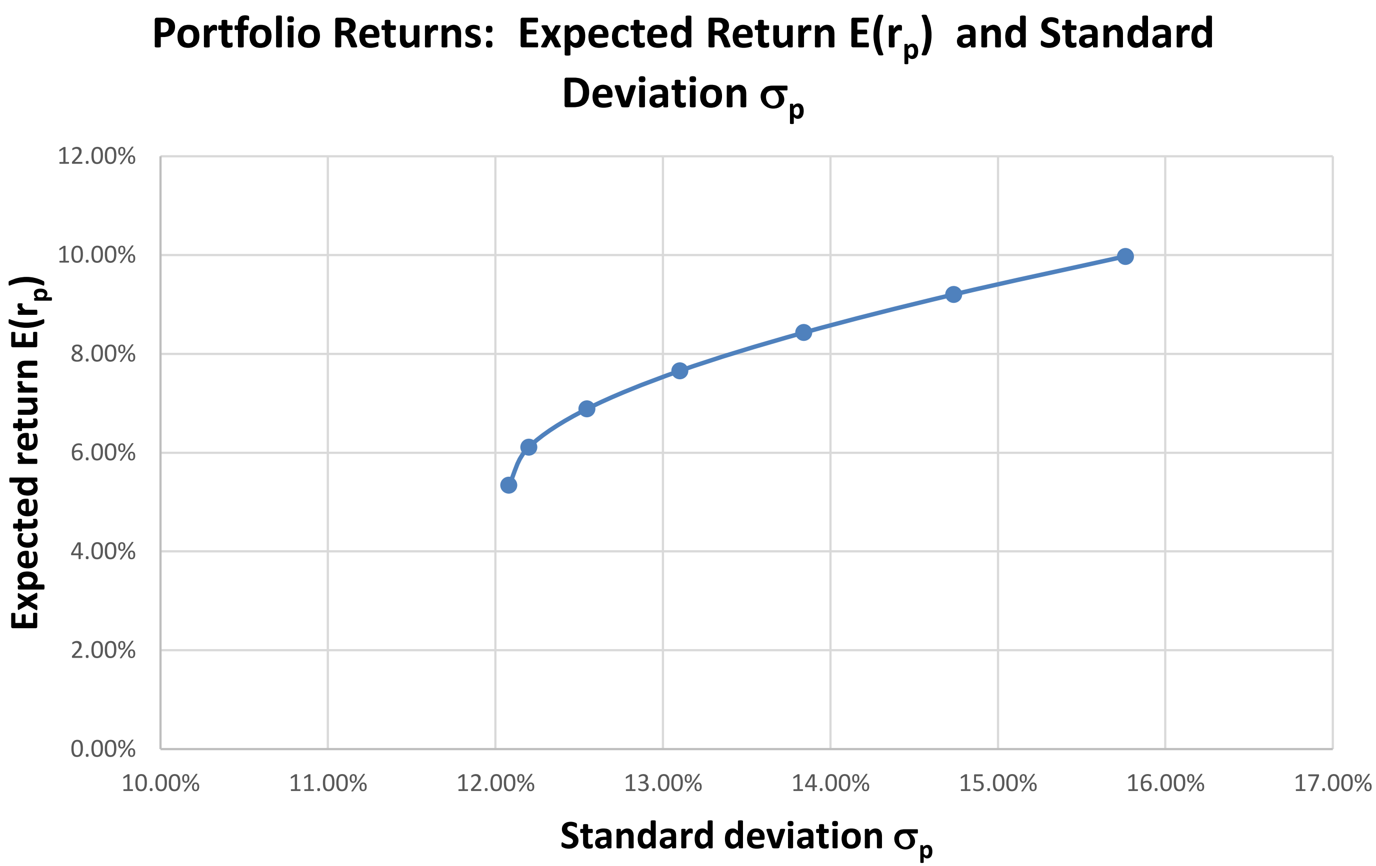
#1. Calculate the requested discrete returns & statistics for the stocks **(good review for the final)**

#2. Populate the requested values for portfolio expected return and standard deviation

#3. Graph the efficient frontier for the two-asset portfolio (return vs. std dev), and find the minimum variance portfolio

#4 Solve for an optimized portfolio

Portfolios – Efficient Frontier

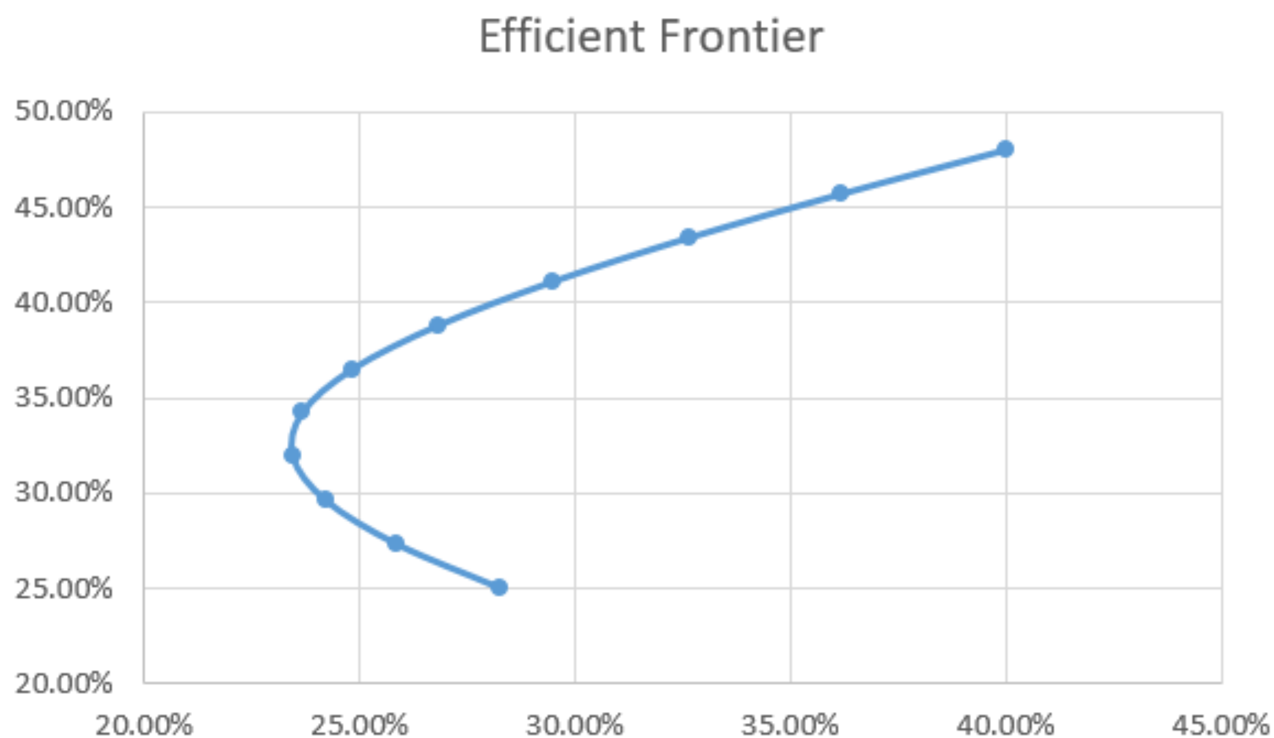


Top part of the curve is the efficient frontier –

Set of those portfolios that offer the highest expected return for a given std. deviation

Efficient Frontier

Open “excel – portfolio optimization”



What if we want a std. deviation of 20%?

Risk Free Asset

Consider a portfolio of two risky assets (stocks A & B) and a risk free asset -

a savings account, government bond, etc.

Let's assume $r_f = 2\%$

https://www.youtube.com/watch?v=6HXoxzZwbCA

Sharpe Ratio (return vs. risk)

$$\frac{(R_p - R_f)}{\sigma_p}$$

Rp = return on portfolio
Rf = the risk-free rate of return
σ = portfolio standard deviation

Portfolio A generates a return of 15%, σ = 8%
Portfolio B generates a return of 12%, σ = 5%

If Rf = 5%: **Sharpe ratio A = 1.25**
Sharpe ratio B = 1.4

Portfolio B was able to generate a higher return on a risk-adjusted basis

Capital Market Line (CML)

By adding a risk-free asset, we generate a linear risk/return trade-off

This line, the CML, provides a better risk/return trade-off than the efficient frontier

We can use Solver to determine weights of portfolio components that maximize Sharpe Ratio

Optimal Investment Combinations

Consider a 50/50 investment in a portfolio of risky assets <u>and</u> the risk free asset that maximizes Sharpe Ratio

What is the expected return and standard deviation of this new portfolio?

(should be lower return/lower risk)

Portfolio Expected Returns

The expected return of a portfolio is the weighted average of the expected returns for each asset in the portfolio:

$$E(R_P) = \sum_{i=1}^{n} w_i E(R_i)$$

w_i = % of portfolio invested in each asset

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Standard Deviation

The variance of the risk-free asset, and the covariance of the risk-free asset with any other asset, are zero!

Std. deviation of a portfolio (comprised of a risky portfolio "*a*" & and risk-free asset) is simply:

$$\boldsymbol{\sigma}_p = w_a \boldsymbol{\sigma}_a$$

Return Variability

Variance:

=var.s()

$$\sigma^2 = \frac{\sum_{i=1}^{n} \left(R_i - \overline{R}\right)^2}{n - 1}$$

(where "n" = number of returns)

Standard Deviation:

=stdev.s()

$$\sigma = \sqrt{\sigma^2}$$

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Covariance

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})}{n - 1}$$

=covariance.s(array1, array2)

Measure the deviation of returns from their average and multiply the deviations